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Project Management 3.1

Estimation Methods

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Learning Outcomes

- By the end of this week, you will be able to:
  - Estimate project costs and times
  - State why estimations are used
  - Discuss the differences between different estimation methods
  - Identify the need for estimations
  - Give relevant estimation examples

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Estimates

- We estimate costs or benefits, because we can’t always know exactly what these will be
- There will be different types of estimates for different types of projects
- Estimates tend to be wide of the mark at the beginning of a project, and closer to the truth as time goes on.

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Types of Costs

- Direct Vs. Indirect
- Recurring Vs. Nonrecurring
- Fixed Vs. Variable
- Normal Vs. Expedited

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Cost Classifications

Costs	Direct	Indirect	Recurring	Non-recurring	Fixed	Variable	Normal	Expedited
Direct Labor	X		X		X		X	
Building Lease		X	X		X		X	
Expedite	X			X		X		X
Material	X		X			X	X	

Derived from Pinto, J. (2019) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

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Cost Estimation

- Ballpark (order of magnitude) ±30%
- Comparative ±15%
- Feasibility ±10%
- Definitive ±5%

Derived from Pinto, J. (2019) Project Management: Achieving Competitive Advantage, 5th Edition, Prentice Hall

Classifying Estimates

This and the next slide informed by: http://en.wikipedia.org/wiki/Cost_estimate

Estimate Class	Name	Purpose	Project Definition Level
Class 5	Order of Magnitude	Screening or Feasibility	50% to 100%
Class 4	Intermediate	Concept Study or Feasibility	30% to 70%
Class 3	Preliminary	Budget, Authorization, or Control	10% to 40%
Class 2	Intermediate	Control or Bid/Tender	1% to 15%
Class 1	Definitive	Check Estimate or Bid/Tender	0% to 2%

Class 1	Class 3	Class 5
Detailed Estimate	Semi-Detailed Estimate	Conceptual Estimate
Final Estimate	Scope Estimate	Pre-Design Estimate
Control Estimate	Sanction Estimate	Preliminary Estimate
As-Bid Estimate		Pre-Budget Estimate
As-Sold Estimate		Evaluation Estimate
CD Estimate	DD Estimate	SD Estimate
		Parametric Estimate
		Rough Order-of-Magnitude (ROM) Estimate
		Very Rough Order-of-Magnitude (VROM) Estimate
		SWAG (Scientific, Wild-A** Guess) Estimate
		PIDOOMA (Pulled It Directly Out Of My A**) Estimate

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COCOMO Constructive Cost Model – Software Development

http://sunset.usc.edu/csse/research/COCOMOII/cocomo_main.html

- Making investment or other financial decisions involving a software development effort
- Setting project budgets and schedules as a basis for planning and control
- Deciding on or negotiating tradeoffs among software cost, schedule, functionality, performance or quality factors
- Making software cost and schedule risk management decisions
- Deciding which parts of a software system to develop, reuse, lease, or purchase
- Making legacy software inventory decisions: what parts to modify, phase out, outsource, etc
- Setting mixed investment strategies to improve organization's software capability, via reuse, tools, process maturity, outsourcing, etc
- Deciding how to implement a process improvement strategy, such as that provided in the SEI CMM

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Ratings

Cost Drivers	Very Low	Low	Nominal	High	Very High	Extra High
Product attributes						
Required software reliability	0.75	0.88	1.00	1.15	1.40	
Size of application database		0.94	1.00	1.08	1.16	
Complexity of the product	0.70	0.85	1.00	1.15	1.30	1.65
Hardware attributes						
Run-time performance constraints			1.00	1.11	1.30	1.66
Memory constraints			1.00	1.06	1.21	1.56
Volatility of the virtual machine environment		0.87	1.00	1.15	1.30	
Required turnabout time		0.87	1.00	1.07	1.15	
Personnel attributes						
Analyst capability	1.46	1.19	1.00	0.86	0.71	
Applications experience	1.29	1.13	1.00	0.91	0.82	
Software engineer capability	1.42	1.17	1.00	0.86	0.70	
Virtual machine experience	1.21	1.10	1.00	0.90		
Programming language experience	1.14	1.07	1.00	0.95		
Project attributes						
Application of software engineering methods	1.24	1.10	1.00	0.91	0.82	
Use of software tools	1.24	1.10	1.00	0.91	0.83	
Required development schedule	1.23	1.08	1.00	1.04	1.10	

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Each of the 15 attributes receives a rating on a six-point scale that ranges from "very low" to "extra high" (in importance or value). An effort multiplier from the table below applies to the rating. The product of all effort multipliers results in an *Effort Adjustment Factor (EAF)*. Typical values for EAF range from 0.9 to 1.4.

The Intermediate Cocomo formula now takes the form:

$$E=a_i(KLoC)^{(b_i)}.EAF$$

where E is the effort applied in person-months, **KLoC** is the estimated number of thousands of delivered lines of code for the project, and **EAF** is the factor calculated above. The coefficient $\mathbf{a_i}$ and the exponent $\mathbf{b_i}$ are given in the next table.
The Development time **D** calculation uses **E** in the same way as in the Basic COCOMO.

Software project	a_i	b_i
Organic	3.2	1.05
Semi-detached	3.0	1.12
Embedded	2.8	1.20

Detailed information is required, but it is still an estimate!

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Problems with Cost Estimation

- ✓ Low initial estimates
- ✓ Unexpected technical difficulties
- ✓ Lack of definition
- ✓ Specification changes
- ✓ External factors
- ✓ Risks turn into issues.

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Cost is not the only thing that can be estimated

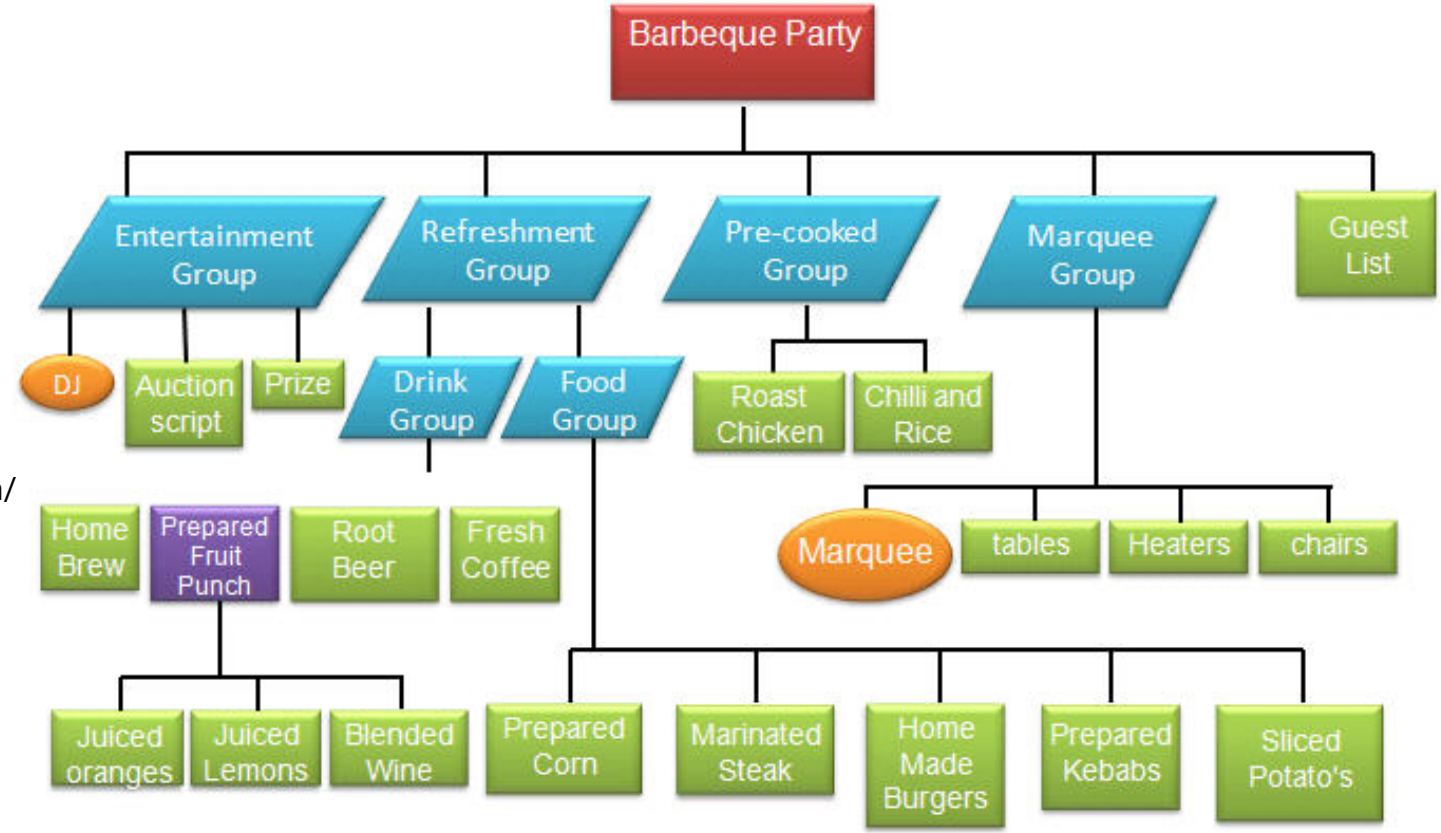
- Time is critical to any project
  - Time estimates are only as accurate as your vision of the future
  - To estimate time, you will need to determine what needs to be done, and take a view on the total time all tasks combined will take
  - You will need to understand the outcomes for your project or any estimate of time will have no accuracy
- Quality is critical to any project
  - Quality estimates are only as good as your requirements planning and specification
  - Quality related costs on a product run between about 10% - 30%, depending on the situation

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Product Breakdown Structure

We can estimate tasks and dependencies

http://www.prince2primer.com/prince2-product-based-planning/



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Work Breakdown Structure

A process that sets a project's scope by **breaking down** its overall **mission into** a cohesive set of synchronous, increasingly **specific tasks**. This helps us to estimate the workload overall

What does WBS accomplish?

- ❖ Echoes project objectives
- ❖ Offers a logical structure
- ❖ Establishes a method of control
- ❖ Communicates project status
- ❖ Improves communication
- ❖ Demonstrates control structure.

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Work Breakdown Structure and Codes

